Quantum No-Cloning Verification REPORT

- TEAM OBSERVERS

1 . Abstract

This project is an integration of the themes Quantum Communication and Quantum Cryptography where we are using simulation to verify the No Cloning Theorem which is a fundamental principle that makes sure of the security of quantum information. By making use of the Qiskit framework and the Qiskit AerSimulator, we are modelling a scenario where an unknown quantum state is prepared by Alice and intercepted by the eavesdropper, Eve. We are implementing a Hadamard (H) and Phase (p) gate sequence to generate a non-orthogonal state, ensuring it can't be perfectly replicated or copied by any linear operator. Our technical implementation shows that anyone who attempts to clone a non-orthogonal state will result in measurable entanglement rather than a perfect copy, creating a physically detectable disturbance.

2 . Introduction and Background

- **Problem Identification:** We are pointing out the vulnerability of information duplication in digital communication where classical data can be copied without getting detected.
- **Motivation:** The real world is in need for unhackable communication and the quantum mechanics offers a unique solution where any attempt to measure or copy alters the data itself. This property is not hold by classical bit.

3 . Methodology

- **State Preparation**: Alice encodes a secret message using Hadamard (H) and Phase (P) gates at pi/4, creating a non-orthogonal quantum state.

- **Simulated Attack**: Eve attempts to copy the message using a CNOT gate, which is what represents a physical eavesdropping attempt.

• **Controlled Execution**: And then we are using Barriers to prevent circuit optimization and run 1,024 shots on the Qiskit Aer simulator for statistical accuracy.

• **Security Audit**: We map outcomes via Histograms and the QSphere to detect the transition from a single qubit to an entangled Bell-state system.

• **Threshold Verification**: The resulting 25% QBER is measured against the 11% Security Threshold to provide physical proof of the breach.

1 . State Preparation(Alice)

- **Qubit q0** : Alice prepares the message qubit.
- **Hadamard (H) Gate** : Puts the qubit into a superposition state.
- **Phase (P) Gate** : Applies at a pi/4 to create a non-orthogonal, unknown quantum state. This is crucial because the No-Cloning Theorem specifically applies to states that are not part of an orthogonal basis.

2 . The Interception (Eve)

- **Qubit q1 :** Represents Eve's target qubit,initially in state |0> .
- **CNOT Gate** : Eve attempts to "clone" Alice's sate by using q0 as the control and q1 as the target.
- **The Quantum Result** : Instead of a copy, the CNOT gate creates entanglement between q0 and q1. This fundamentally alters the states of the original qubit (q0), making the attack detectable.

3 . Measurement & Analysis

- **Standard Measurement** : Both qubits are measured and the results are stored in the 2-bit classical register.
- **Simulation** : The circuit is executed using Qiskit Aer to gather probability distributions.
- **Metric** : The Quantum Bit Error Rate (QBER) is calculated by comparing Alice's intended transmission with the measured reslut of qo after Eve's interference.

4 . Result and Discussion

The circuit was executed on the **Qiskit Aer simulator** and the resulting data is visualized through **QSphere** and **Histogram**. Unlike classical ”tapping” which has 0% error, Our quantum simulation result in 25% QBER,significantly exceeding the 11% security Threshold.

5 . Conclusion and References

Practical Feasibility:The result shows that quantum channel are highly feasible for detecting unauthorized access.

Limitation: Currently it is used for single qubit interception; multi-qubit system may require more complex error correction.

Future work: We can apply this protocol to protect critical identity operations, such as aadhaar load stabilization.

References:

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